A distinct-support factorial-product collision

A counterexample to Aivazidis-Loukaki-Müller, Problem 5.6

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Abstract. Two disjoint sets of integers greater than one, of cardinalities nine and seven, have equal products of f(t) = Gamma(t+1)/t^t. An explicit prime-valuation certificate proves the identity exactly. This answers negatively the arithmetic question in published Problem 5.6 of On the common transversal probability. A separate Lean certificate proves the literal real Gamma statement.

1. Statement and source scope

Aivazidis, Loukaki and Müller [1, Problem 5.6, printed p. 1281] ask whether equality of products of f(t) at distinct integers greater than one forces equal lengths and equality after rearrangement. The published question has no equal-cardinality, equal-sum or group-realization premise. Its preprint number is Problem 12.6. We use the published numbering throughout.

Theorem. Let A = {3,5,6,7,8,14,16,20,36} and B = {4,10,12,15,18,21,32}. These sets are nonempty and disjoint, every entry exceeds one, and

product over A of Gamma(t+1)/t^t = product over B of Gamma(t+1)/t^t.

Since |A| = 9 and |B| = 7, the proposed implication in Problem 5.6 is false.

The witness satisfies both natural readings of distinctness: distinct entries within each list, and all entries distinct across both lists. No minimality, general classification, asymptotic theorem or group realization is asserted.

2. Exact finite proof

For a prime p and a positive rational r, write v_p(r) for its prime exponent. Counting the multiples of p, p², and so on in t! gives

v_p(t!/t^t) = sum over j >= 1 of floor(t/p^j) - t v_p(t). (1)

Applying (1) gives the following complete list of nonzero prime totals. All primes greater than 36 have valuation zero on both sides.

p	A total	B total
2	-123	-123
3	-37	-37
5	-4	-4
7	-8	-8
11	6	6
13	5	5
17	3	3
19	2	2
23	1	1
29	1	1
31	1	1

Unique factorization of positive rationals therefore identifies both products with

(11^6 · 13^5 · 17^3 · 19^2 · 23 · 29 · 31) / (2^123 · 3^37 · 5^4 · 7^8).

At every positive integer t, Gamma(t+1) = t!, so this is the literal source identity. The sixteen displayed entries are pairwise distinct and greater than one. Their two lengths are unequal, which contradicts the length conjunct of the proposed conclusion. This completes the proof.

3. Auditable certificate and formal endpoint

For independent hand checking, the next table gives every individual exponent from (1). Its first nine rows form A and its last seven form B. Thus the preceding proof depends only on finite integer arithmetic and unique factorization, rather than floating-point approximation or an unbounded search.

t	2	3	5	7	11	13	17	19	23	29	31
3	1	-2	0	0	0	0	0	0	0	0	0
5	3	1	-4	0	0	0	0	0	0	0	0
6	-2	-4	1	0	0	0	0	0	0	0	0
7	4	2	1	-6	0	0	0	0	0	0	0
8	-17	2	1	1	0	0	0	0	0	0	0
14	-3	5	2	-12	1	1	0	0	0	0	0
16	-49	6	3	2	1	1	0	0	0	0	0
20	-22	8	-16	2	1	1	1	1	0	0	0
36	-38	-55	8	5	3	2	2	1	1	1	1
4	-5	1	0	0	0	0	0	0	0	0	0
10	-2	4	-8	1	0	0	0	0	0	0	0
12	-14	-7	2	1	1	0	0	0	0	0	0
15	11	-9	-12	2	1	1	0	0	0	0	0
18	-2	-28	3	2	1	1	1	0	0	0	0
21	18	-12	4	-18	1	1	1	1	0	0	0
32	-129	14	7	4	2	2	1	1	1	1	1

The companion certificate.py recomputes each of the 176 entries by (1) and by separate factorization of t! and t^t. It also checks exact rational multiplication and integer cross multiplication. These checks support the displayed certificate; they are not numerical evidence substituted for a proof.

The unchanged mathematical source FactorialProduct.lean defines qValue(t) = t!/t^t in the rationals and f(x) = Real.Gamma(x+1)/x^x in the reals, with real exponentiation. gamma_dictionary uses Real.Gamma_nat_eq_factorial and Real.rpow_natCast. Induction transports rational list products to the real products. Nodup, disjointness, nonemptiness and cardinalities are proved, not assumed.

The endpoint integer_source_statement_negation unconditionally negates the universal statement for integer lists with Nodup, entries greater than one and equal real f-products, whose conclusion is equal lengths and List.Perm. The endpoint disjoint_nonempty_counterexample proves the stronger finite-set witness. The natural-list and finite-set formulations are auxiliary versions of the same refutation.

The source-only replay compiles one mathematical module and inventories all 31 actual module-origin declarations using import all, including private/generated declarations. It records actual elaborated types, origins, transitive axioms and safety flags. Only propext, Classical.choice and Quot.sound occur. The mathematical compilelog is empty; no sorry, admit, additional axiom, unsafe definition or native_decide is used. Audit tooling is separate from the mathematical module.

4. Relation to the source theorem

Theorem 4.6 of [1, pp. 1263-1265] requires one entire side to consist of distinct primes. Both A and B contain composites, so neither orientation satisfies its hypothesis. Its largest-prime and majorization argument depends on

that prime-side restriction. In particular, the positive high-prime valuations in our table can occur on both sides; the zero-valuation step on the source's prime side is unavailable. Our identity neither contradicts that theorem nor extends it by its proof.

5. Status, prior mathematics and limits

This is a compact finite counterexample, with no claim of unusual technical depth. A complete alternate existence argument using smooth-number restriction, classical concentration and pigeonhole was found in independent preliminary review. That full argument and all objections are retained in preliminary-review.md, alongside the complete prior-review.md. The alternate proof itself is a serious reason to regard the result as a routine application; independent final review may reject substantive eligibility.

Briefly, for $L = \log N$ and T the integers up to N with all prime factors at most N/L^2, a prime-count estimate gives $|T| = N - o(N)$. Small-prime valuation coordinates cost $\exp(o(N))$ possibilities. For larger primes, the factorial valuations are $\text{floor}(t/p)$, bounded by L^2. Concentration confines most subset sums to widths $\sqrt{N}$ times powers of $\log N$. Their joint box has $\exp((1.25506/2 + o(1))N)$ possibilities, fewer than the $2^{(N-o(N))}$ subsets. Two subsets collide; cancellation gives disjoint supports. This supplies the necessary coefficients in {-1,0,1}; a bare integer kernel would not.

Candela-Helfgott [2, Proposition 2.1] explicitly uses concentration boxes for distinct subset sums. The mechanism is established prior mathematics. Its small 1-span hypothesis alone does not provide the factorial smooth restriction and two-scale valuation estimates: prime generators do not 1-span columns with unbounded coefficients. Whether that remaining assembly is substantive is a judgment, not a theorem. No inspected earlier body was found to supply the assembled factorial collision; this is bounded historical coverage, not a worldwide priority claim.

The retained prior review records exact registry coverage through Omega 0d4df45e499f5e2d97471c506c555c18fd44920c and OpenAI 7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a on 29 October 2026, and the access limits. Jevtic's adjacent integral-vector codebook paper, DOI 10.1137/S0895480194265623, remains unread after HTTP 403. It is a historical limitation, not evidence of absence. No fresh literature clearance is claimed by this manuscript preparation.

Root's current dispositions are scoped substantive PASS and CLEAR_BOUNDED, with the foregoing dissent intact. Independent final review and acceptance remain pending; credit is zero. The author line follows the campaign's Research manuscript convention and invents no human author. Source PDF, text, request receipts, exact pins, complete reviews, failure logs, root decisions and replay instructions accompany the packet. The original mathematical source bytes are unchanged.

References

[1] S. Aivazidis, M. Loukaki and T. W. Müller, On the common transversal probability, Journal of Group Theory 27 (2024), 1233-1285. DOI 10.1515/jgth-2024-0030. Published source retained from https://d-nb.info/1346963460/34; Problem 5.6 on printed p. 1281.

[2] P. Candela and H. A. Helfgott, On the dimension of additive sets, Acta Arithmetica 167 (2015), 91-100. DOI 10.4064/aa167-1-5. Proposition 2.1 and its concentration-box proof; exact retained excerpt provenance is in the packet.

[3] J. B. Rosser and L. Schoenfeld, Approximate formulas for some functions of prime numbers, Illinois Journal of Mathematics 6 (1962), 64-94. Section 3, Corollary 1, (3.6), as used in the complete alternate proof retained in the preliminary review.